Proctor's Odd Symplectic Group

What can be averaged, exact replacements, and computational verification

Proof and reproducibility note

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Outcome. The proposed normalized Haar expectation for the full complex odd symplectic group is not defined: the group contains both a vector group and $\mathbb{C}^\times$. Three valid replacements were proved and tested. Exact constant-term calculations verify the two compact formulas through total degree six. The full algebraic invariant series is exactly 1. For finite fields, four matrix groups of orders 24, 432, 12,000, and 11,520 were enumerated in dimensions three and five; all 38 tested coefficients agree between direct permutation-matrix averages and the proposed cycle formula. Selected coefficients also agree with explicit orbit counts.

1 The obstruction comes before computation

Let $V = W \oplus L$ over $\mathbb{C}$, where $\dim W = 2n$, $L = \mathbb{C}e_0$, and ω is a nondegenerate alternating form on W. Extend it by

$$\Omega((w,s),(w',s')) = \omega(w,w'), \qquad \operatorname{rad}\Omega = L.$$

Consider the full stabilizer

$$G = \{g \in GL(V) : \Omega(gv, gw) = \Omega(v,w)\}.$$

Because the radical is intrinsic, $gL = L$. In the decomposition $W \oplus L$, every element therefore has the form

$$g = \begin{pmatrix} A & 0 \\ \ell & a \end{pmatrix}, \qquad A \in \mathrm{Sp}(W), \quad \ell \in W^*, \quad a \in \mathbb{C}^\times. \tag{1}$$

Conversely, every matrix in (1) preserves Ω. Thus

$$G \cong W^* \rtimes \big(\mathrm{Sp}_{2n}(\mathbb{C}) \times \mathbb{C}^\times\big). \tag{2}$$

The vector group W^* is the unipotent radical, so G is nonreductive; both W^* and $\mathbb{C}^\times$ are noncompact. A locally compact group has a normalized Haar probability exactly when it is compact. Consequently

$$\frac{1}{\mathrm{vol}(G)} \int_G f(g)\, dg$$

is not a defined normalization. This is a mathematical obstruction, not a numerical convergence problem. A finite box cutoff would depend on coordinates and cutoff shape and is not a group-invariant substitute.

There is a useful convention warning. Some authors impose $g(e_0) = e_0$. We write this subgroup as

$$G^1 = W^* \rtimes \mathrm{Sp}_{2n}(\mathbb{C}), \tag{3}$$

which is obtained by setting $a = 1$. The full stabilizer G and G^1 give different answers below and should not be conflated.

2 A structural identity and what it sees

Block triangularity gives, for every scalar t,

$$\det(I - tg) = (1 - at)\det(I - tA), \qquad \mathrm{Spec}(g) = \mathrm{Spec}(A) \cup \{a\}. \tag{4}$$

The shear coordinate ℓ can change Jordan structure but is invisible to the characteristic polynomial. Therefore any probability measure ν for which the coefficientwise moments exist satisfies

$$\int_G \prod_i \det(I - t_i g)^{-1}\, d\nu(g) = \int_{\mathrm{Sp}_{2n}(\mathbb{C})\times\mathbb{C}^\times} \prod_i \frac{\det(I - t_i A)^{-1}}{1 - at_i}\, d\bar{\nu}(A, a), \tag{5}$$

where $\bar{\nu}$ is the pushforward to the Levi quotient. This gives a useful diagnostic for heat-kernel or user-chosen noncompact probabilities: the answer depends on the chosen measure, but never on its conditional distribution in the unipotent coordinate.

The code tests (4) modulo q on deterministic samples from every constructed finite group. It is important that this structural test is separate from the permutation-representation test in Section 5.

3 Two compact replacements

A maximal compact subgroup of the full stabilizer is

$$K = \mathrm{Sp}(n) \times U(1), \qquad V|_K = W \oplus \chi,$$

where $\chi(z) = z$ on the radical line. Write

$$\mathcal{S}_{H,X}(\mathbf{t}) = \int_H \prod_i \det(I - t_i h|_X)^{-1}\, dh$$

for a compact group with normalized Haar probability. Direct-sum multiplicativity and product Haar measure give

$$\mathcal{S}_{K,V} = \mathcal{S}_{\mathrm{Sp}(n),W}\mathcal{S}_{U(1),\chi}.$$

Since

$$\prod_i (1 - zt_i)^{-1} = \sum_{d\geq 0} z^d h_d(\mathbf{t}), \qquad \int_{U(1)} z^d\, dz = \delta_{d0},$$

we obtain the exact identity

$$\boxed{\mathcal{S}_{K,V} = \mathcal{S}_{\mathrm{Sp}(n),W}.} \tag{6}$$

The radical line disappears because it has a nontrivial $U(1)$ weight.

For the fixed-radical group G^1, a maximal compact subgroup is $\mathrm{Sp}(n)$ and $V|_{\mathrm{Sp}(n)} = W \oplus \mathbf{1}$. The extra eigenvalue is now 1, so

$$\boxed{\mathcal{S}_{\mathrm{Sp}(n),W\oplus\mathbf{1}} = \mathcal{S}_{\mathrm{Sp}(n),W} \prod_i (1 - t_i)^{-1}.} \tag{7}$$

Exact rank-one check

For $n=1$, restrict to the maximal torus of $SU(2)=\mathrm{Sp}(1)$ with weights z, z^{-1}. Weyl integration is the exact constant-term operation

$$\int_{SU(2)} f = \mathrm{CT}_z\big((1-z^2)f(z)\big).$$

The verification represents every complete homogeneous character as a Laurent polynomial. It extracts both the z and $U(1)$ constant terms and tests all 30 partitions through total degree six. Representative results are:

τ	K direct	$\mathrm{Sp}(1), W$	G^1 compact direct	formula (7)
$()$	1	1	1	1
(1)	0	0	1	1
(2)	0	0	1	1
$(1,1)$	1	1	2	2
$(2,1)$	0	0	2	2
$(2,2)$	1	1	3	3

All 30 comparisons pass exactly, with integer arithmetic.

4 The full algebraic invariant series

Even though Haar averaging fails, the covariant invariant series is meaningful:

$$\mathcal{S}^{\mathrm{inv}}_{G,V} = \sum_{\tau} \dim \left(\bigotimes_j \mathrm{Sym}^{\tau_j} V \right)^G m_\tau. \tag{8}$$

Theorem. For the full stabilizer G in (1),

$$\boxed{\mathcal{S}^{\mathrm{inv}}_{G,V} = 1.} \tag{9}$$

Proof. Embed $\bigotimes_j \mathrm{Sym}^{\tau_j} V$ into $V^{\otimes d}$, where $d=|\tau|$. The subgroup $\mathrm{diag}(I_W, a)$ acts with weight a^r on tensors having r radical-line factors. Its invariant subspace is thus contained in $W^{\otimes d}$.

For $\ell \in W^*$, differentiate the action of

$$u_\ell = \begin{pmatrix} I_W & 0 \\ \ell & 1 \end{pmatrix}.$$

On $x \in W^{\otimes d}$ the derivative is a sum of d contractions, one for each tensor position, and the output of the jth contraction has its unique e_0 in position j. These outputs lie in distinct direct summands. Hence invariance forces every positional contraction against every ℓ to vanish. Already at the first position, the contractions by a basis of W^* recover all coordinates of x, so $x=0$. Thus there are no positive-degree invariants. □

This proof gives a compact computational certificate: after the torus weight filter, stack the first-position contractions against a basis of W^*. The result is a coordinate permutation of rank $(2n)^d$. The implementation records full rank for $n=1,2$ and $1 \le d \le 4$ (eight checks), so every recorded kernel has dimension zero. This certificate is stronger and cheaper than building large Reynolds matrices for a nonreductive group.

5 Finite odd symplectic groups and a genuine complex representation

Let $E = W \oplus L$ over a prime field $\mathbb{F}_q$. The same block proof gives

$$G^{\mathrm{odd}}_{2n+1}(q) \cong W^* \rtimes \left(\mathrm{Sp}_{2n}(q) \times \mathbb{F}_q^\times\right), \tag{10}$$

and hence

$$\begin{aligned} |G^{\mathrm{odd}}_{2n+1}(q)| &= q^{2n}(q-1)|\,\mathrm{Sp}_{2n}(q)|, \\ |\,\mathrm{Sp}_{2n}(q)| &= q^{n^2}\prod_{j=1}^{n}(q^{2j}-1). \end{aligned} \tag{11}$$

Matrices with entries in $\mathbb{F}_q$ are not automatically complex matrices. We therefore use the honest complex permutation representation on $S = \mathbb{P}(E)$. If P_g is its permutation matrix and $c_r(g)$ is the number of r-cycles, then

$$\det(I - tP_g)^{-1} = \prod_{r\geq 1}(1-t^r)^{-c_r(g)}. \tag{12}$$

Consequently the proposed finite formula is

$$\boxed{\mathcal{S}_{G(q),\mathbb{C}[S]}(\mathbf{t}) = \frac{1}{|G(q)|}\sum_{g\in G(q)}\prod_i\prod_{r\geq 1}(1-t_i^r)^{-c_r(g)}.} \tag{13}$$

For a partition τ, Burnside's lemma also gives

$$[m_\tau]\mathcal{S}_{G(q),\mathbb{C}[S]} = \#\left(G(q)\backslash\prod_j \mathrm{MSet}_{\tau_j}(S)\right). \tag{14}$$

There are exactly two orbits on $\mathbb{P}(E)$: the radical point and all nonradical points. The latter assertion follows because $\mathrm{Sp}(W)$ is transitive on nonzero vectors of W, while $\ell(w)$ adjusts the radical coordinate. Therefore

$$\boxed{[m_{(1)}]\mathcal{S}_{G(q),\mathbb{C}[\mathbb{P}(E)]} = 2.} \tag{15}$$

6 Independent verification algorithms

For each (n, q), the program performs the following steps.

1. Enumerate every $2n \times 2n$ matrix A satisfying $A^\mathsf{T} JA = J$, then every triple (A, ℓ, a) in (10). Check the observed orders against (11), preservation of the degenerate form, and deterministic closure samples.

2. Enumerate normalized representatives of every projective point and construct the permutation P_g for every group matrix.

3. **Direct matrix route:** compute $p_k = \mathrm{tr}(P_g^k)$ from explicit permutation powers and use Newton's recurrence

$$dh_d = \sum_{k=1}^{d} p_k h_{d-k}.$$

   Average $\prod_j h_{\tau_j}$ over all matrices.

4. **Cycle route:** independently find the cycles and expand the right side of (12) by dynamic programming. Average the resulting products and compare every coefficient with the direct route.
5. **Orbit route:** in the two smaller cases, enumerate the orbits on multisets for $\tau = (1), (2), (1,1)$ and compare with (14).

The direct route does not call the cycle-product routine. This separation matters: otherwise (13) would only be tested against a rearrangement of itself.

7 Exact results

n	q	$\dim E$	$\lvert \mathrm{Sp}_{2n}(q)\rvert$	$\lvert G(q)\rvert$	$\lvert\mathbb{P}(E)\rvert$	tested degrees
1	2	3	6	24	7	all partitions through 4
1	3	3	24	432	13	all partitions through 4
1	5	3	120	12,000	31	all partitions through 3
2	2	5	720	11,520	31	all partitions through 3

All four form, order, radical-point, determinant, closure, and coefficient checks pass. In total, 38 matrix-average coefficients match 38 cycle-formula coefficients exactly. Selected values illustrate both stability and genuine dimension dependence:

(n,q)	$m_{(1)}$	$m_{(2)}$	$m_{(1,1)}$	$m_{(3)}$	$m_{(2,1)}$	$m_{(1,1,1)}$
$(1,2)$	2	5	6	11	18	25
$(1,3)$	2	5	6	12	19	28
$(1,5)$	2	5	6	14	22	30
$(2,2)$	2	6	7	19	31	43

For $(n,q) = (1,2)$ and $(1,3)$, explicit orbit enumeration gives respectively 2, 5, and 6 orbits for $\tau = (1), (2), (1,1)$, exactly matching both averaging routes.

8 Interpretation, limitations, and new tools

The results separate four notions that are easy to blur:

- The full complex-group Haar expression is undefined, and no computation can validate it without first choosing a new measure.
- Maximal-compact averaging is canonical up to conjugacy, but it forgets the unipotent radical. For the full stabilizer it also removes the radical line by $U(1)$ weight; for G^1 it retains that line as a trivial summand.
- The full covariant invariant series is well-defined but collapses to 1 in the defining representation. Interesting odd symplectic representation theory therefore requires other modules, mixed V, V^* constructions, or polynomial-function invariants.
- Finite-field permutation actions give exact, nontrivial, reproducible Λ-distributions, but they are different representations of different finite groups and should not be presented as complex matrix realizations of the defining modular matrices.

Two reusable tools emerge. First, the *Levi visibility test* (4)–(5) immediately decides which nonreductive coordinates an eigenvalue statistic can detect. Second, the *torus-filter plus derivative certificate* proves invariant vanishing using small full-rank contraction maps instead of a nonexistent Reynolds average.

9 Reproducibility

From the repository root, run

```
.venv/bin/python \
  for_this_guy/proctor_odd_symplectic_group/verification.py
```

and open the interactive report with

```
.venv/bin/marimo edit \
  for_this_guy/proctor_odd_symplectic_group/notebook.py
```

The verification uses only exact integer and rational arithmetic. Exhaustive group enumeration is intentionally limited to prime fields and to the four displayed cases; extending to larger ranks should replace all-matrix search by standard symplectic generators.

References

[1] S. Okada, *A bialternant formula for odd symplectic characters and its application*, 2019, https://arxiv.org/abs/1905.12964.

[2] W. Burnside, *Theory of Groups of Finite Order*, second edition, 1911.

[3] A. Haar, *Der Massbegriff in der Theorie der kontinuierlichen Gruppen*, Annals of Mathematics 34 (1933), 147–169.